Week 6 Progress Writeup

1. **Each team member's contribution:**

Shreya - Signed off on propensity-weighted Qini as the official evaluation metric, after confirming it already combines the common-support trimming I'd been planning separately with inverse-propensity weighting. No need to build a 2nd fix. Investigated Claudia's concern that gas/kiosk and misc transaction lines inflate spend-based features. Confirmed it's real (8.4% of all dollars) but quantified that it doesn't change Model 3's conclusions (88.5% of households land in the same spend tier either way), and left the actual fix to her since it's shared feature code. Closed the gap between how the model trains and how it's judged, then ran 5 real causal-inference approaches against it: a propensity-weighted T-learner, an X-learner built for our 12:1 treated/control imbalance, a doubly-robust/AIPW learner, and entropy balancing. Caught and fixed real bugs in 2 of them (an optimizer divergence in entropy balancing, a pickling failure when saving the X-learner for production scoring) before trusting either result. X-learner came out ahead, a real gain on open campaigns, still positive but not statistically proven on the 1 true held-out check, and is now Model 3's model in use, with the old T-learner kept as a permanent comparison. Two other candidates that looked like wins on open data were caught failing that same held-out check, which is the system working as intended, not a setback. Added an E-value sensitivity analysis (1.35) so "no hidden confounders" is a measured number instead of an assumption, a confounder-as-feature column, and a standalone "sure things" classifier (AUC 0.675) now wired into the team's shared Lead Agent. Tested that Lead Agent directly and gave Pranav concrete feedback on it. Published the full candidate comparison and reasoning to the repo, and cleaned up leftover mentor-attribution language across Model 3's files.

John - This week I followed up on our mentors' feedback for Model 1 by testing which features add signal beyond coupon use. In the Modeling Agent, I added coupon use alone as the bar to beat plus "coupon use plus one feature group" candidates and found that no single existing group clearly improved on coupon use. I then built model1_feature_search.py to screen 38 candidate features on the open test campaigns only and found the useful signal in more detailed coupon habits, coupon use on each campaign's own products and categories, more detailed redemption history, and demographics. On the feature engineering side, I added 18 of these columns to Feature Agent 2A. I also added a logistic regression on log-scaled values as a model option. With these changes, the Evaluation Agent marked three alternatives as reliably better than the baseline on the open campaigns for the first time. On integration, I rebased the work onto teammates' updates, connected the new options to the lead agent, and merged everything as PR #29. I presented these results to our mentors and took their feedback on feature importance and alternative model types as next steps.

Anna - This week I followed up on the Evaluation Agent's flag that Model 2's median line predicted totals at only 80% of actual spend, outside the ±15% the Decision Lab needs. Before looking at any results, I fixed three rules for replacing the model in use (totals within 15%, beat usual spend, beat the current line by at least $0.50 per household), then tested 10 candidates through the Modeling and Evaluation Agents in two rounds: totals fixes (a median line rescaled to totals, a Tweedie regression) and additional feature groups added to usual spend. None passed all three rules; the closest fixed totals (0.92) but was only $0.29 better, and the extra feature groups were reliably worse, which points to data size, not features, as Model 2's limit. I left the locked test campaigns unread, merged the candidates, unit tests, a full comparison report

and a design-doc update as PR #30, and presented the results to our mentors. Next, I'll run a small side experiment on a broader Model 2 target.

Pranav - This week I changed how our agents fit together. Thinking it through from a data scientist's side, I realised that the person using the system won't have built it. They won't know what the Data, Feature, Modeling and Evaluation Agents each do, and they shouldn't have to. So I designed and built a lead agent. It's the only agent the data scientist talks to, and it runs the other four behind the scenes. Ask it to run Model 1 and it checks for new data, rebuilds only what changed, retrains when a campaign has finished, grades the result and explains it in plain words. Plain code does every calculation, and the AI only explains what the code measured, so no number is made up. A retrained model replaces the one in use only if it passes every check, and switching to a different kind of model is always a person's call. Known issues are already rules, so only something new stops a run. When that happens, it asks instead of guessing, and saves the answer as a rule with who decided and why, so the same question never comes up twice. Our data is fixed, so I built a replay that runs the system as of any day to simulate new data arriving. It caught a real surprise: a new "VIDEO" department on day 595, which turned out to be beer filed under the wrong department. After I answered, it retrained on 20 campaigns and replaced the previous version automatically, reaching 55.3% of redeemers in its top 20%, 12.1 points above the "redeemed before" rule (+7.8 to +16.9). On the full data, it found that John's new logistic regression beats Model 1 by 3.8 points, and asked me instead of switching. I presented it with a live demo at Friday's mentor meeting. Once the Decision Lab is built, we'll deploy the whole system on Google Cloud, with the chat running on Gemini.

Zhiyan - This week I completed four Data Agent follow-ups from the advisor meeting. I investigated the 4,451 lines with a quantity but no sales value and found that store promotions, not coupons, explain 83% of them (coupons only 1%), with no link to campaigns. I added a `quantity_imputed` column (product median, else 1) for the 7 probable missing-quantity lines, leaving the original quantity untouched; this is merged into main. I documented the evidence for the two typo rules: the 9 lines marked as probable typos fall inside a normal range measured from the data, and the 24 lines sent to a person do not. Finally, I found that gas and miscellaneous lines are 1.1% of lines but 8.3% of dollars and sit inside every Model 2 and 3 spend feature, so I built an off-by-default switch to exclude them (keeping campaign coupon products in); whether they count as spend is now a decision for the advisors.

2. Plan for upcoming week

Next week, we plan to finish the models and the agentic system and deploy them on Google Cloud, then start designing the Decision Lab. For Model 1, we'll follow up on the mentors' feedback on feature importance and other model types, and check logistic regression on the held-out test campaigns before deciding whether to switch to it. For Model 2, we'll keep the least-squares usual-spend model and run a small side experiment on a broader target. For Model 3, we'll make the confounder feature specific to each household's shop and test the X-learner again. On the agentic side, we'll deploy the lead agent on Google Cloud, with the engine checking for new data on a daily timer and the chat running on Gemini. We'll then begin designing the Decision Lab: what a marketer needs to see, and what each model has to hand over to it. We're hoping to show our mentors an early version where a mock marketer asks the system questions.